<table>
<tr><td>Module Code</td><td>CM 4450</td><td>Module Title</td><td colspan="5">Computational Bayesian Statistics</td></tr>
<tr><td>Credits</td><td>2.5</td><td rowspan="2">Hours/Week</td><td>Lectures</td><td>2</td><td rowspan="2">Pre-requisites</td><td rowspan="2" colspan="2">CM 2420</td></tr>
<tr><td>GPA/NGPA</td><td>GPA</td><td>Lab/Assignments</td><td>3/2</td></tr>
<tr><td colspan="2">Module Objectives</td><td colspan="6">To provide students with in-depth understanding of theory and concepts in Bayesian model construction and inference.</td></tr>
<tr><td colspan="2" rowspan="7">Learning Outcomes</td><td colspan="6">At the completion of this module, students should be able to;</td></tr>
<tr><td>LO1</td><td colspan="5">Explain in detail the Bayesian framework for data analysis, its flexibility and demonstrate when the Bayesian approach can be beneficial.</td></tr>
<tr><td>LO2</td><td colspan="5">Distinguish Bayesian and frequentist statistical paradigms.</td></tr>
<tr><td>LO3</td><td colspan="5">Develop any problem from a Bayesian perspective.</td></tr>
<tr><td>LO4</td><td colspan="5">Use simulation algorithms to perform a Bayesian analysis of more complex Models.</td></tr>
<tr><td>LO5</td><td colspan="5">Perform Bayesian prediction and decision making.</td></tr>
<tr><td>LO6</td><td colspan="5">Perform Bayesian model inference.</td></tr>
<tr><td colspan="2">Outline Syllabus</td><td colspan="6">● The Bayesian paradigm for statistical inference
● Bayesian computation
○ Markov Chain Monte Carlo, Hamiltonian Monte Carlo (HMC), No-U-Turn Sampler (NUTS), Variational Bayes
● Bayesian modelling
○ Linear Models, Mixture models, Latent Dirichlet Allocation, State space models, Gaussian processes, Smoothing with basis expansions
● Bayesian Learning
○ Bayesian Neural Networks, Bayesian Additive Regression Trees (BART)</td></tr>
<tr><td colspan="2">Method of Assessment</td><td colspan="6">● Final Examination- 70% (Closed Book)
● Continuous assessments - 30%</td></tr>
<tr><td colspan="2">Recommended text</td><td colspan="6">1. Bishop, C. M. (2006). Pattern recognition and machine learning. Springer
2. Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). Bayesian data analysis. CRC press.
3. Turkman, M. A. A., Paulino, C. D., and Müller, P. (2019). Computational Bayesian Statistics: An Introduction (Vol. 11). Cambridge University Press.</td></tr>
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